Ultimate Guide (Types, Application & Pro Tips)

Cheese is far more than a simple topping; it is a complex ingredient whose composition directly dictates the outcome of your pizza. From the way it melts and stretches to how it browns and releases oils, every characteristic is governed by variables like fat content, moisture levels, and application methods. This article provides a comprehensive look at the world of pizza cheese – led by the mozzarella – and offering practical tips for selection and application

Pizza Cheese: An Introduction

Cheese is one of the core ingredients in pizza. Along with the dough and tomato sauce, it plays a major role in shaping the flavor, texture, and overall appearance of the pizza.

Choosing a cheese for your pizza might seem straightforward – after all, pizza = mozzarella, right? Not quite. Several characteristics of cheese directly influence how it behaves on a pizza, including:

- The type of cheese
- Its fat content
- Its moisture content
- Its flavor
- The way it's cut
- The amount used

Each of these factors, on their own and together, affects various aspects of the pizza, including:

Flavor Impact: Stronger cheeses like Parmesan or sharp Cheddar will contribute a more pronounced flavor, while milder cheeses such as mozzarella have a more neutral presence. Some cheeses are best added after baking, such as ricotta or burrata, to bring an additional layer of flavor.

Baking Behavior: The shape and size of the cheese – whether sliced, cubed, grated, thick or thin – affect how it melts, browns, and releases fat during baking. Different cutting methods give us better control over the cheese's behavior in the oven.

Visual Effect: The type of cheese, how it's cut, and how it bakes all influence the final appearance of the pizza. Cheese that fully melts and separates from its fat tends to blend with the sauce, creating a cohesive red/orange cheese-sauce layer. In contrast, thicker cuts that bake more slowly may retain their original color and stay more distinct. For example, fresh mozzarella often remains bright white after baking, while cheddar typically develops a yellow to orange hue.

Texture: Cheese also has a direct impact on texture, contributing to whether the pizza feels soft, stretchy (the cheese), rubbery, oily, etc.

Fat Content: The fat content in cheese significantly affects its flavor, texture, and baking performance. Higher-fat cheeses tend to melt more evenly and taste richer, while lower-fat cheeses may brown faster or resist melting.

Much like tomato sauce, the cheese you choose and how you use it helps define your pizza's character. Different cheeses, applied in different ways and cuts, can create a completely unique result.

While **mozzarella** is by far the most popular and widely used cheese in pizza-making (and we'll cover it in detail later), it's only one option among many.

How Cheese Affects Baking and the Final Result (With Illustrative Photos)

The type of cheese you use, whether mozzarella (of any kind) or other cheeses, and how you apply it, has a major impact on the baking process and, ultimately, the finished pizza.

Cheese influences pizza in several key areas, each affecting how it melts, browns, and contributes to flavor, texture, and visual appeal:

- The type and quality of cheese
- The degree to which the cheese is baked
- The fat content
- The moisture content
- The way the cheese is cut and applied (grated, cubed, striped, sliced, torn by hand)
- The amount of cheese used

Each of these aspects will be covered in detail in the sections that follow. You'll also find images accompanying each section to visually demonstrate how these variables influence cheese behavior [assuming all other things being equal*].

*By “all other things being equal,” I mean that the results shown reflect changes in a single factor (the section discussed), while everything else – cheese type, quantity, cut, and baking conditions, remains the same.

The Impact of Cheese Type and Quality

There are many types of cheese, some more suitable for pizza than others, each bringing distinct characteristics that influence the final result:

- Varying levels of moisture and fat (more on this later)
- Unique flavor profiles (nutty, sweet, sharp, sour, milky, etc.)
- Different melting behaviors during baking (affected by both type and quality)
- Varying textures after baking (creamy, rubbery, soft, stretchy – also influenced by type and quality)

Experimenting with different cheeses lets us create diverse flavor and texture combinations on pizza. Mozzarella, for instance, is the most widely used cheese for pizza thanks to its excellent **melting and browning** behavior, along with its signature **stretch**.

Beyond the type of cheese, **quality** plays a crucial role in baking performance. Higher-quality cheeses typically resist fat separation and over-browning, resulting in better melt, more consistent texture, and a superior overall outcome.

An example of how cheese type and quality affect texture and appearance: Low-moisture mozzarella (the red-tinted slices) vs. generic yellow cheese (yellowish in color with a pimpled, uneven surface)

The Degree of Cheese Baking

Let’s first look at how varying degrees of baking change the characteristics of the cheese on pizza.

When Cheese Is Baked **Less**:

- It remains more distinct from the sauce, retaining its original color and sitting on the pizza as a separate “layer.”
- It browns minimally or not at all.
- It releases less fat.
- It tends to stretch more after baking.

- If applied in a thick layer, it can develop a rubbery, elastic texture once the pizza cools, unlike fully melted cheese that has blended into the sauce.

When Cheese Is Baked **More**:

- It blends more thoroughly with the sauce, creating a uniform, red/orange-tinted cheese-sauce layer (assuming tomato sauce is used).
- It browns faster.
- It releases more fat.
- Its texture holds up better after cooling, due to its fusion with the sauce.

The next sections will detail **the main factors that affect how much and how quickly cheese bakes**: the cut of the cheese, the type used, and its fat and moisture content.

In addition to these cheese-related variables, other factors also influence baking:

1. **Baking temperature**: Higher temperatures accelerate melting and browning.
2. **Proximity to heat source**: Cheese closer to direct heat (like an oven element or open flame) bakes more quickly.
3. **Cheese temperature**: Cold or frozen cheese will bake more slowly.
4. **Timing of addition**: Adding cheese partway through baking reduces its exposure to heat. The later it's added, the less it bakes.

At the end of the article, you'll find practical tips for controlling the rate and degree of cheese baking.

Fully baked cheese: Fat has separated, browning has begun, and the cheese formed a unified layer with the sauce (low-moisture mozzarella, 25% fat)

The Role of Fat Content in Flavor and Texture

First and foremost, just like with meat, **fat = flavor**. In general, the higher the fat content of a cheese, the more flavorful and pronounced its taste will be.

But fat affects more than just flavor. It plays a critical role in how cheese behaves during baking, and influences the final texture of both the cheese and the pizza. Fat content affects how cheese melts, stretches, and whether it turns out soft and tender, or tough, chewy, and rubbery.

Fat acts as a plasticizer – it lubricates the protein matrix (casein), making it more flexible and less rigid when heated. More fat = less protein per gram, which means less structure forms during melting and baking, resulting in a **softer, more tender**

texture.

Lower-fat cheeses contain more protein and water, which can form a tighter network upon heating, making the cheese rubbery or chewy if overcooked. During baking, fat melts before proteins, helping the cheese to flow and pool rather than dry out.
In General, **Higher-Fat** Cheeses:

- Are more resistant to browning and drying out during baking.
- Yield a softer, more tender texture after baking, with better stretch.

Lower-Fat Cheeses**:**

- Brown more quickly.
- Tend to develop a rubbery or chewy texture after baking.

Part-skim, low-moisture mozzarella (21% fat)

Higher fat content also increases the likelihood of **fat separation** during baking, often resulting in a visible layer of melted fat ("grease") pooling in the center of the pizza. This liquid fat usually mixes with the sauce and creates an orange-hued layer, as seen in the example below.

Example of fat separation: note the prominent, orange "grease" layer on the pizza (A blend of whole milk, low-moisture mozzarella, 25% fat, and Gouda, 28% fat)

Is fat separation a good or bad thing? That depends on personal preference. On the one hand, the melted fat mixed with the sauce is full of flavor; On the other hand, too much fat separation – common with cheeses containing 25% fat or more – can make the pizza overly greasy. Some people love it, while others don't.

Historically, a classic hallmark of New York style pizza was the stream of orange grease dripping from a hot slice down your hand and elbow.

What's the recommended fat content for pizza? For most pizzas (aside from Neapolitan pizza baked in a wood-fired oven), the ideal cheese fat content is between **22% and 30%** for optimal melt, stretch, and flavor. Neapolitan pizza, which typically uses **fresh mozzarella** [from cow's milk], usually falls in the **16%–19%** fat range.

Later in the article, you'll find a complete table listing various cheeses along with their fat content.

How Moisture Content Dictates Melting and Browning

The moisture (water) content of cheese plays a key role in how it behaves during baking.

In general, cheeses with **higher moisture content**:

- Brown less during baking, since browning only begins once the surface has dried out.
- Spread more during baking, covering a wider area as the moisture seeps out.
- Result in a softer, creamier, and more "wet" texture after baking.

On the other hand, cheeses with **lower moisture content**:

- Brown more easily and more intensely.
- Spread less during baking, maintaining a more compact coverage.
- Develop a firmer texture after baking.

A classic example of a **high-moisture** cheese is **fresh mozzarella**, which contains about 60–70% moisture. It's most commonly used on Neapolitan pizzas, creating the iconic look of bright white "islands" of cheese.

Fresh mozzarella on Neapolitan pizza: Retains its white color and creates a soft, creamy "puddle"-like texture and appearance

In contrast, **very low-moisture cheeses**, such as **Parmesan and Pecorino**, with moisture levels around 30%, are not suitable as the primary cheese on pizza. They brown too quickly, melt poorly, and are best used as finishing cheeses or blended with other cheeses, rather than used as the main cheese.

Later in the article, you'll find a complete table listing various cheeses along with their moisture content.

Influence of Cut and Application Techniques

The way cheese – regardless of type – is cut and applied, has a significant effect on how it melts, browns, and behaves during baking. Cheese can be used in various forms, including:

- Grated
- Cubed
- Cut into strips or sticks
- Sliced

- Torn by hand (from/into chunks or slices)

Each of these forms can **vary in size or thickness**. For example:

- Grated cheese can range from fine flakes to thick, long shreds
- Cubes can be small or large
- Strips can be thin and short or thick and long
- Slices can be paper-thin or quite thick
- Torn pieces can be roughly torn into small bits or large chunks

How Thickness Affects Melting and Baking

In general, the **thicker** the cheese – whether grated, cubed, sliced, or torn – the **longer** it will take to:

- Melt
- Brown
- Release fat
- Blend with the sauce

For example, cheese grated into thick shreds will melt more slowly than the same cheese finely grated. Likewise, cubes, strips, or thick slices will bake more slowly than thinner versions of the same cut.

Summary:

- **Thinner/lighter** pieces melt and brown faster, release fat more quickly, and blend into the sauce more easily.
- **Thicker/larger** pieces take longer to undergo all of the above changes.

How Cheese Shape Affects Coverage

Beyond melting behavior, the way cheese is cut also affects how evenly and thoroughly it covers the pizza base.

In general, **the smaller the cut**, the **greater and more even the coverage** (assuming the same amount of cheese is used).

Here is how different cheese shapes affect coverage, listed from the greatest coverage to the least coverage:

- **Shredded cheese** provides the most coverage. The finer it's shredded, the more coverage it provides.

- **Small cubes** offer fairly good coverage, though less than shredded cheese. The larger the cubes, the more gaps will remain, and the smaller the coverage.
- **Sliced cheese** (round or square) can also provide good coverage if the slices overlap, but this usually requires a larger quantity of cheese (unless the slices are very thin, in which case they may bake too quickly).
- **Strips or chunks** offer the least coverage, especially when cut thick.

The Amount of Cheese Used

The amount of cheese applied to a pizza significantly affects how it bakes and the final result.

Assuming all other factors being equal (type of cheese, baking method, cut size, etc.):

- **More cheese** takes longer to melt and brown, resulting in a slower and more gradual baking.
- **Less cheese** melts and browns more quickly, leading to faster baking.

Below are visual examples showing how different amounts of cheese can impact the pizza's appearance and texture:

Types of Cheese, Their Properties, and How to Use Them on Pizza

Classifying Cheeses for Pizza: A General Introduction

While mozzarella is the most commonly used cheese on pizza, many other cheeses also work beautifully, either on their own or in combination with mozzarella. Because mozzarella has a mild, neutral flavor, blending it with other cheeses opens the door to creative experimentation with flavor and texture.

For pizza making, cheeses can be grouped into four broad categories:

1. Fresh/soft cheeses
2. Semi-soft cheeses
3. Semi-hard cheeses
4. Hard/firm cheeses

For simplicity and practicality, the classifications used here don't strictly follow 'official' cheese taxonomy. For instance, mozzarella (whether fresh or low-moisture) is technically considered a semi-soft or soft cheese; However, its actual texture is

often firmer than that of typical soft cheeses, so in this context, I categorized it as a semi-hard cheese.

The Milk Used to Make the Cheese (Cow's Milk vs. Sheep/Goat Milk)

In addition to texture-based classifications, cheese can also be categorized by the **type of milk** used to make it. Most cheeses are made from either **cow's milk** or **sheep/goat milk**.

For simplicity, from this point on, the term "sheep's milk" will refer to both sheep's and goat's milk.

The primary differences between cow's milk cheeses and sheep's milk cheeses is in **flavor, texture, and nutritional composition**:

- **Cow's milk cheeses** generally have a milder, more neutral flavor with a soft, milky aroma.
- **Sheep's milk cheeses** typically have a more intense, tangy, and sour flavor, often accompanied by a stronger aroma. Sheep's milk cheeses are also typically more expensive than cow's milk cheeses

Sheep and goat milk contain more fat and protein than cow's milk, resulting in cheeses that are **richer in flavor and creamier in texture**. Classic examples include **feta**, **pecorino**, and **manchego** – each has a bold, distinctive flavor.

Cheese Aging (Aged vs. Young Cheeses)

Cheese aging, or ripening, is the process of allowing cheese to mature over time, from a few days to several years, under controlled conditions of temperature and humidity. Aging enhances the cheese's **flavor, texture, and aroma**.

Aged cheeses are also known as *mature cheeses*; In contrast, young cheeses, often referred to as "mild" cheeses, are aged for only a short period or not at all.

During aging, several changes occur in the cheese:

- Moisture evaporates
- Proteins and fats break down
- Flavors become more concentrated and complex
- The texture becomes firmer, sometimes even crumbly

These changes are why long-aged cheeses often don't melt well and are typically used for grating or as a seasoning, rather than as a primary pizza topping.

In general, **the longer a cheese is aged**, the **stronger its flavor**, the **firmer its texture**, and the **higher its fat and salt concentration** (due to moisture loss, which concentrates the remaining solids).

Examples:

- **Parmesan and Pecorino** are aged from **3 to 36 months**, developing a hard, grainy, or crumbly texture and a sharp, complex flavor.
- **Cheddar or Gouda**, when young, are **softer and milder**; as they age, they develop **deeper, sharper flavors and firmer textures**.
- **Mozzarella** (bot fresh and low moisture), which is **unaged**, has a delicate, creamy, milky taste.

Comprehensive Guide to Cheese Categories and Applications

1. Fresh and Soft Cheeses

Examples: *Burrata, mascarpone, ricotta, cream cheese, quark, tvorog, and any creamy, spreadable cheese.*

These cheeses are typically unaged and have very high moisture content, delicate flavors, and soft, spreadable textures. Because of their high moisture and sensitivity to heat, they are not suitable for baking applications.

How to Use:

In order to preserve their fresh flavor and creamy texture, fresh/soft cheeses should be added **after baking** . They can be spooned on in dollops, piped with a pastry bag, or gently spread over the pizza after it has slightly cooled.

2. Semi-Soft Cheeses

Examples: *Blue cheeses (Gorgonzola, Roquefort, etc.), Brie, Camembert, goat cheeses like chèvre or Bûche.*

These cheeses are creamier than semi-hard cheeses, but firmer than soft cheeses. Their flavors range from mild and buttery to sharp and tangy, depending on the specific variety and its age.

How to Use:

Semi-soft cheeses melt partially (depending on the specific variety) and retain much of their flavor during baking. They're ideal as accent cheeses, paired with a milder base like mozzarella, and can be used sparingly alongside the main cheese **before baking**.

Semi-soft cheeses can also be added **after baking** for a bolder, fresher finish, much like soft or fresh cheeses.

3. Semi-Hard Cheeses

Examples: *Mozzarella (all types), yellow cheeses [generic], Emmental, young Gouda, Provolone, young Asiago, Monterey Jack, Taleggio, young/mild Cheddar, Caciocavallo, Sulguni, Scamorza, Muenster, Gruyère, Fontina, Brick.*

This is the main or most important category for pizza-making. Semi-hard cheeses offer the ideal baking qualities: they have excellent melting properties, brown moderately, and provide the balanced stretch, chew, and tenderness that define great pizza cheese.

How to Use:

These cheeses can be used **on their own** or **blended** with others (especially mozzarella), to create the desired flavor and texture. All the principles outlined in previous sections regarding how cheese affects baking and the final result, apply primarily to cheeses in this category.

4. Hard and Firm Cheeses

Cheeses in this category fall into two groups: hard, aged cheeses (like Parmesan and similar varieties that undergo long aging), and firm, salty cheeses with relatively high moisture content (like feta). While their characteristics differ, their usage on pizza is largely similar.

Examples [Hard aged cheeses]: *Parmesan, Pecorino, Grana Padano, Manchego, aged Asiago*
Examples [Salty firm cheeses]: *Feta, Sirene, brynza.*

Cheeses in this category are used **more for seasoning or topping** than as the main cheese, due to their limited melting ability.

Hard aged cheeses (e.g. Parmesan, Grana Padano, Pecorino, Manchego) are rich in flavor and ideal for grating over pizza, **either before or after baking**.

Firm, moist cheeses (e.g. feta) can be used as toppings **before baking**, as they hold enough moisture and fat to avoid drying out, or **after baking**, to preserve their fresh texture and flavor.

How to Use:

Hard cheeses like Parmesan can be added either **before or after baking**, with each method offering different results:

- **Before baking (grated):** Produces a more delicate, mellow flavor due to heat exposure and melting. Some of the cheese's original sharpness or complexity is softened, but it blends well with the sauce and other cheeses. If exposed directly to heat (e.g., grated on top of the cheese with no barrier), it may brown too much. This method is ideal when you want the cheese to subtly season the pizza.
- **After baking (grated or thinly sliced):** Delivers a stronger, more concentrated flavor, as the cheese remains untouched by heat. The cheese stands out more and adds distinct bursts of seasoning. This method preserves the cheese's umami and is perfect for a bold finishing touch.

You can also **combine both methods** to enjoy the best of both worlds: some flavor baked into the pizza, and some left fresh on top.

Firm, salty cheeses like feta can be used similarly:

- **Before baking (in cubes or slices):** Leads to a drier, firmer texture due to moisture loss, and reduces the fresh flavor. The result is a more intense, salty cheese with a slightly chewy texture.
- **After baking:** Preserves the cheese's original creamy texture and tangy flavor.

Summary Table: Cheese Varieties and Their Properties

Below is a summary table of common cheeses cheeses, including their flavor profiles and key characteristics.

Note: This is only a partial list of popular cheeses – many more can be used on pizza, as outlined throughout this article.

Fresh/Soft Cheeses:

	Milk Source	Fat (per 100g)	Moisture	Flavor Profile	Properties & Notes
Burrata	Cow	22-25%	60-65%	Creamy, milky, rich, slightly tangy, delicate	Doesn't melt well (not suitable for baking); Add only after baking.
Mascarpone	Cow	35-40%	~60%	Neutral, buttery, slightly sweet	Doesn't melt. best used as post-bake topping or white pizza base.
Ricotta	Cow/Sheep/Mixed	10-13%	~70%	Milky, mildly sweet, delicate	Doesn't melt (but softens in baking). Use post-bake or as a white pizza base.
Cream Cheese	Cow	5-35%	~55%	Creamy texture, sweet-tangy flavor	Doesn't melt (but softens in baking). Best added after baking.
Quark	Cow	10-15%	65-75%	Mild tangy,	More suited for filling than

	Milk Source	Fat (per 100g)	Moisture	Flavor Profile	Properties & Notes
				yogurt-like	topping; add only after baking.
Tvorog	Cow	9-18%	65-70%	Tangy, slightly sweet	More suited for filling than topping; add only after baking.

Semi-Soft Cheeses:

	Milk Source	Fat (per 100g)	Moisture	Flavor Profile	Properties & Notes
Blue Cheeses (Gorgonzola, Roquefort, etc.)	Cow/Sheep	28-35%	40-50%	Very strong, pungent (varies by cheese type)	Melts decently. Can be mixed with other cheeses. Use in small amounts to avoid overpowering flavor.
Brie	Cow	25-30%	~50%	Creamy, earthy, mushroomy	Partially melts into creamy texture.
Camembert	Cow	22-28%	~48%	Like Brie, but more intense	Melts well, stronger flavor. Good in cheese blends.

	Milk Source	Fat (per 100g)	Moisture	Flavor Profile	Properties & Notes
Chèvre / Bûche (Goat cheese)	Goat	20-24%	~45%	Tangy, pronounced goat-milk flavor	Softens but doesn't fully melt; adds a dominant flavor.

Semi-Hard Cheeses:

	Milk Source	Fat (per 100g)	Moisture	Flavor Profile	Properties & Notes
Fresh Mozzarella	Cow/Buffalo	16-19%	60-70%	Mild, milky, neutral	Melts into a wet/creamy texture, minimal browning. Ideal for Neapolitan pizza.
Low-Moisture Mozzarella	Cow	21-27%	45-52%	Mild, milky, more concentrated than fresh mozzarella	Excellent melt and stretch, lightly browns. Ideal for most pizza types.
Scamorza	Cow	25-27%	~45%	Similar to mozzarella, but more concentrated	Melts well. Good mozzarella alternative.

	Milk Source	Fat (per 100g)	Moisture	Flavor Profile	Properties & Notes
Sulguni	Cow/Buffalo/Mixed	22-28%	45-52%	Similar to mozzarella, more intense and salty	"Georgian mozzarella". firmer and denser. Good mozzarella alternative.
Provolone	Cow	25-28%	~44%	Similar to mozzarella, sharper	Great melt and stretch. Great alone or blended.
Emmental	Cow	28-31%	~38%	Nutty, mild	Partial melt. Good in blends.
Kashkaval	Cow/Sheep/Mixed	25-38%	~42%	Piquant, salty, rich, nutty	Melts well. The cheddar of the Balkans.
Taleggio	Cow	27-30%	~48%	Creamy, slightly salty, fruity, sharp	Melts excellently. Strong flavor.
Young Gouda	Cow	28-30%	~42%	Sweet, nutty	Melts well. Mild flavor. Good alone or in blends.
Monterey Jack	Cow	30-32%	~43%	Subtle, buttery, creamy, slightly sweet,	Excellent meltability. Great for blending.

	Milk Source	Fat (per 100g)	Moisture	Flavor Profile	Properties & Notes
				slightly tangy	
Young/Mild Cheddar	Cow	33-35%	~38%	Buttery, creamy	Not the best melt. Significant oil separation when used alone. Best blended.
Fontina	Cow	27-31%	~40%	Creamy, nutty, mild	Great melt. Balanced, pleasant flavor. Good for white pizza or blends.
Gruyère	Cow	32-35%	~36%	Nutty, salty, quite strong	Great browning and melting properties. Adds flavor depth.
Brick	Cow	27-33%	~43%	Creamy, nutty, slightly sweet	Excellent melt. Distinct flavor. Traditionally used in Detroit-style pizza.

	Milk Source	Fat (per 100g)	Moisture	Flavor Profile	Properties & Notes
Muenster	Cow	27-30%	~43%	Mild, creamy	Melts well, decent browning. Pairs well with mozzarella.

Hard Cheeses:

	Milk Source	Fat (per 100g)	Moisture	Flavor Profile	Properties & Notes
Parmesan	Cow	25-38%	~30%	Sharp, salty, nutty, rich, umami	Doesn't melt well (too dry). Use grated before/after baking.
Pecorino	Sheep	27-30%	~32%	Sharp, salty, strong sheep-milk flavor	Doesn't melt well. Use grated before/after baking.
Grana Padano	Cow	28-30%	~32%	Similar to Parmesan but milder	Doesn't melt well. Use grated before/after baking.
Aged Asiago	Cow	28-30%	~30%	Sharp, nutty, sweet, creamy, peppery	Doesn't melt well. Use grated before/after baking.

	Milk Source	Fat (per 100g)	Moisture	Flavor Profile	Properties & Notes
Manchego	Sheep	27-30%	~40%	Nutty, bold flavor	Partially melts. Adds strong aroma. Good grated or blended.
Feta	Sheep	5-22%	~55%	Salty, tangy	Doesn't melt. Adds saltiness. Can be added crumbled/cubed before or after baking.
Halloumi	Sheep/Goat/Cow	25-27%	43-47%	Salty, tangy, milky	Very high melting point, not suitable for pizza.
Bulgarian Cheese (*Sirene*)	Cow/Sheep	5-22%	~55%	Sharper and saltier than feta	Doesn't melt. Adds saltiness. Can be added crumbled/cubed before or after baking.

Mozzarella: The Queen of Pizza Cheeses

In the following sections, we'll cover everything one needs to know about mozzarella in general, and its use in pizza applications in particular.

Understanding Mozzarella's Dominance in Pizza

Mozzarella belongs to the **Pasta Filata** cheese family (Italian for "stretched curd"), a name that reflects its production method. Depending on the type, mozzarella can be technically classified as either semi-soft (in the case of low-moisture mozzarella) or soft (for fresh mozzarella).

Mozzarella is made by acidifying milk (cow's or water buffalo's), and adding rennet to form curds. The curds are cut, heated, and stirred to expel whey, then left to rest and acidify further. Once they reach the right acidity (pH), the curds are stretched in hot water (a process called *pasta filata*) until smooth and elastic.

At this point, mozzarella can take one of two forms:

- **Fresh mozzarella** is shaped into balls and packed in brine or whey to retain its high moisture content (60–70%).
- **Low-moisture (LM) mozzarella** is dried to reach a low(er) moisture content (45–52%) and a firmer texture.

In the next section, we'll go into the differences between fresh and low-moisture, as well as the distinctions between cow's milk and buffalo milk versions (for fresh mozzarella).

Across all types, mozzarella is characterized by a **mild, milky, neutral flavor**. It's not meant to take center stage, but rather to complement and support other ingredients. Its subtle flavor makes it ideal as a background player in the overall flavor profile of the pizza.

So if not for flavor, why is mozzarella the default cheese for pizza? Beyond tradition, it comes down to performance: **mozzarella melts exceptionally well, browns evenly, and develops a stretchy, tender-chewy texture after baking** – these qualities are what make it the queen of pizza cheeses.

For many, its delicate flavor is also an asset, allowing the sauce, dough, and toppings to shine, while still contributing some richness and body.

Another key reason for mozzarella's popularity as pizza cheese: it is generally **much more affordable** (e.g. cheaper) than most other cheeses suited for pizza, making it both a practical and functional choice.

Comparing Fresh vs. Low-Moisture Mozzarella

Mozzarella can be categorized along three key dimensions:

- **Milk source:** Cow's milk ("fior di latte") vs. buffalo's milk ("fior di bufala")
- **Moisture content:** Fresh mozzarella vs. low-moisture (LM) mozzarella
- **Fat content:** Full-fat/whole milk (WM) vs. part-skim (PS)

The primary/only difference between whole milk and part-skim mozzarella is **fat content** – whole milk mozzarella contains more fat than part-skim.

The most common types of mozzarella used in pizza making (and in general) are:

- **Low-moisture, whole milk** (LMWM) mozzarella from cow's milk [Fat content: 24–27% per 100g] – the traditional mozzarella for most types of pizza.
- **Low-moisture, part-skim** (LMPS) mozzarella from cow's milk [Fat content: 21–23% per 100g] – a lower-fat variation.
- **Fresh mozzarella from whole cow's milk** [Fat content: 16–19% per 100g]
- **Fresh mozzarella from whole buffalo's milk** [Fat content: 22–25% per 100g]

These types of mozzarella are rare (if they exist at all):
– Low-moisture mozzarella from buffalo's milk
– Fresh mozzarella from part-skim buffalo's/cow's milk

Fresh Mozzarella ("Mozzarella Fresca")

Fresh mozzarella, or *mozzarella fresca* in Italian ("fresca" meaning "fresh"), has a **high moisture content**, typically between 60–70%. It is most commonly formed into balls weighing 80–100 grams, and packed in liquid, which acts as a natural preservative.

This liquid is usually either whey (a byproduct of cheesemaking) or a light brine. It helps preserve the cheese's soft texture, prevents it from drying out or becoming rubbery, and slightly extends its [already] short shelf life.

When stored properly (see storage tips at the end of the article), fresh mozzarella typically keeps for 1–2 weeks past its sell-by date if unopened, and 3–7 days once opened.

Nutritionally, fresh mozzarella is lower in fat than low-moisture mozzarella, with a fat content of around **16–19%,** and contains about half the sodium (salt) – typically **150–250 mg** (per 100 grams).

Fresh mozzarella is the traditional cheese used on **Neapolitan pizza**. It's also popular in salads (such as Caprese) and in a wide range of cold sandwiches and dishes.

Interestingly, the mozzarella of choice for many pizzerias in Naples, including Da Michele and other renowned pizzerias, isn't traditional fresh mozzarella, but a special semi-moist variety called **Fior d'Agerola**. This is a fresh yet drier cow's milk mozzarella (*fior di latte*), with a moisture content of around 56%.

Thanks to its lower moisture, Fior d'Agerola doesn't require draining (see the section on using fresh mozzarella for more details), and is easy to cut, shred, or slice, making it ideal for professional use.

Fresh Mozzarella from Buffalo Milk (Fior di Bufala Mozzarella)

Buffalo mozzarella, which is almost always sold as fresh mozzarella (no low-moisture versions), is considered a premium product and is significantly more expensive than cow's milk fresh mozzarella.

Buffalo milk is richer than cow's milk – it's whiter, thicker, and contains about 8% fat compared to about 4% in cow's milk. It also has more protein (around 4.5% vs. 3.3%), about 40% more calcium, and 25–40% less cholesterol.

Like cow's milk mozzarella, buffalo mozzarella is sold in liquid (whey or brine) to preserve its texture and freshness.

Thanks to its higher fat and protein content, buffalo mozzarella has a **slightly softer, more elastic texture**, and a **richer, creamier flavor** compared to cow's milk mozzarella.

In terms of usage, buffalo mozzarella is treated just like fresh cow's milk mozzarella when used on pizza.

Low-Moisture Mozzarella ("Mozzarella Block/Loaf")

Low-moisture mozzarella contains significantly less water than fresh mozzarella, with a typical moisture content of **45% to 52%**. It is usually sold in block form or as a long "sausage"-shaped loaf.

Low-moisture mozzarella is typically dried within hours of production and then immediately packaged.. When properly stored (see storage tips at the end of the article), it has a shelf life of 3 to 4 weeks in the refrigerator if unopened, and around 2 weeks once opened.

The flavor of low-moisture mozzarella is **more concentrated and noticeably saltier** than that of fresh mozzarella – this is due to its lower water content, which increases the concentration of fat, protein, and salt. It typically contains **21% to 27%** fat and **400 to 700 mg** of sodium (per 100 grams).

Low-moisture mozzarella is the most common and widely recognized pizza cheese around the world. It is the go-to choice for almost every style of pizza outside of Neapolitan (that said, it can still be used for Neapolitan pizza – see tips at the end of the article).

When and How to Use Each Type of Mozzarella

Fresh Mozzarella

Fresh mozzarella is especially well-suited for **baking at very high temperatures**, particularly for Neapolitan pizzas baked in wood-fired ovens. There are two main reasons for this:

- **Tradition:** Fresh mozzarella is the original cheese used in Naples and remains the standard. Using fresh mozzarella preserves authenticity and closely replicates the original in terms of flavor (very delicate), texture ("puddles" or "islands" on the pizza), and appearance (cheese that retains its bright white color).
- **Baking behavior:** Due to its high moisture content, fresh mozzarella is highly resistant to browning, making it ideal for high-heat baking. In contrast, low-moisture mozzarella (especially part-skim) tends to brown too quickly at such temperatures, resulting in a less desirable texture and appearance.

Fresh mozzarella on a Neapolitan pizza baked in a wood-fired oven

This doesn't mean low-moisture mozzarella can't be used for Neapolitan-style pizza. It certainly can, and with excellent results, if used correctly (see the tips at the end of the article).

Key characteristics of fresh mozzarella:

- Has a very delicate, milky flavor (compared to low-moisture mozzarella)
- Contains less fat (16–19%, or 22–25% for buffalo mozzarella)
- Highly resistant to browning during baking
- Melts faster than low-moisture mozzarella
- Retains a bright white color during and after baking
- Does not release fat during baking
- Creates a creamy, moist texture after baking
- Floats above the sauce rather than blending into it
- Does not hold toppings well – melted fresh mozzarella is slippery and may cause toppings to slide
- Has low stretchability after baking compared to low-moisture mozzarella
- Loses creaminess upon cooling, can become rubbery/spongy

- Does not reheat well

In addition to high-temperature baking, fresh mozzarella can also be used where its unique visual and textural qualities are desired. For example, you can add pieces of fresh mozzarella on top of low-moisture mozzarella to create contrast – white "islands" of fresh mozzarella sitting above a melted, browned base that has blended with the sauce.

Low-Moisture Mozzarella

Low-moisture mozzarella is the **standard and ideal choice for nearly all types of pizza** (other than Neapolitan), especially those baked at lower temperatures (below ~660°F / 350°C), such as New York style, Sicilian, Detroit style and more.

If you're aiming for the classic pizza appearance, flavor, and texture (excluding Neapolitan) – low-moisture mozzarella is the right choice.

Key characteristics of low-moisture mozzarella:

- More concentrated and salty flavor due to lower moisture and brief aging
- Higher fat content than fresh mozzarella (21–27%)
- Browns well during baking
- Releases fat during baking
- Melts more slowly than fresh mozzarella
- Blends seamlessly with the sauce during baking
- Ideal for achieving a classic pizza look
- High stretchability after baking
- Excellent reheating performance

Quality Variations Between Dairies and Manufacturers

Different dairies use **varying production methods** and **source milk from cows raised and treated under different conditions**. These factors directly affect the quality of the mozzarella, especially in the context of pizza, where proper melting, browning, stretching, and texture are essential.

Mozzarella from different dairies also differs in **fat content**, typically ranging from 21–27% for low-moisture mozzarella and 16–19% for fresh [cow's milk] mozzarella. As discussed earlier, fat content plays a significant role in both flavor and baking performance.

In most cases, the primary differences among mozzarella from different producers relate to **baking performance** – how the cheese melts, stretches, and behaves after baking.

Lower-quality mozzarella might:

- Brown too quickly or unevenly
- Melt poorly or unevenly
- Release excessive fat during baking
- Develop an unpleasant texture (plastic-like, rubbery, overly oily, or dry)
- Exhibit poor reheating performance

Flavor differences* between brands are generally **minor**. Mozzarella has a mild, neutral flavor, and in the context of pizza, with sauce, dough, and toppings, subtle differences in taste are rarely noticeable. A clear exception is **smoked** mozzarella, which has a distinctive smoky flavor.

* Assuming cheeses of similar type and fat content (e.g., comparing two low-moisture mozzarellas from different diaries, both with 25% fat). A 25% fat cheese will always taste richer and more flavorful than a 21% fat version, regardless of the dairy, due to fat's role in flavor.

In summary, differences between mozzarella from various dairies include:

- How it melts, browns, and releases fat during baking
- The final texture after baking (e.g., stretchy, soft, rubbery, oily, dry)
- Reheating performance

- Subtle flavor differences

Summary Table: Differences Between Fresh and Low-Moisture Mozzarella

Below is a table summarizing the differences between fresh and low-moisture

mozzarella:

Feature	Fresh Mozzarella	Low-Moisture Mozzarella
Moisture Content	60–70%	45–52%
Fat Content	16–19% (22-25% for buffalo mozzarella)	21–27%
Form	Balls packed in brine or whey	Block or loaf
Shelf Life	3-7 days (opened), 1–2 weeks (unopened)	Up to 2 weeks (opened), 3–4 weeks (unopened)
Sodium Content	150–250 mg per 100g	400–700 mg per 100g
Texture	Soft, moist, delicate	Firm, drier
Flavor	Mild, milky, creamy	Richer, saltier, more concentrated
Melting/Baking Behavior	Releases a lot of moisture, forms puddles, minimal (if any) fat separation, does not brown	Melts uniformly, Integrates with the sauce and forms a cohesive cheese-sauce layer, browns, releases fat
Post-Baking Behavior	Soft/tender, creamy, moist; forms distinctive white “islands”; stays separate from sauce; low stretchability	Structured, more chewy/elastic, oily/greasy, high stretchability (“cheese pull”)
Texture After Cooling	Loses creaminess, becomes rubbery or watery, can feel spongy	Firms up but retains chew and structure
Reheating Performance	Doesn’t reheat well, separates into curd and water	Good reheating performance, especially when using whole-milk (higher-fat) and high-quality cheese

Feature	Fresh Mozzarella	Low-Moisture Mozzarella
Common Uses	Neapolitan pizza, salads, cold sandwiches/dishes	Most non-Neapolitan pizzas; not ideal for very high-temperature baking (e.g., wood-fired ovens)
How to Use	Cannot be shredded (too moist), must be torn or cut into cubes/strips. Requires draining	Can be shredded, sliced, or cubed easily

Tips for Using Cheese

In the following sections, you'll find practical tips and answers to common questions about using cheese on pizza.

Practical Guide to Sourcing and Buying Pizza Cheese

In general, it's best to buy cheese – of any type – as a **whole block**, rather than pre-sliced or pre-shredded. If possible, opt for cheese sold at a deli counter or from a specialty cheese shop, where it's cut fresh to order.

Otherwise, **buying a block of cheese from the refrigerated section and shredding or slicing it yourself at home is the recommended choice**. This not only ensures you're getting higher-quality cheese with minimal additives, resulting in **better performance** for pizza, but also gives you **full control** over the shape and thickness of the cut.

Pre-shredded cheese products, which are widely available in most supermarkets, typically contain **anti-caking agents**, such as starches or cellulose (E460), and sometimes preservatives. These additives interfere with how the cheese melts and browns, and alter its final texture – **making pre-shredded cheese a subpar choice for pizza**.

The anti-caking agent is essentially a powder that coats the cheese and prevents it from sticking to itself inside the package, allowing for convenient use. These agents burn quickly during baking, causing the cheese to brown too fast, and affecting its melt and final texture. Therefore, it is advisable to avoid them.

Pre-packaged cheeses (especially pre-shredded ones) are often **more expensive** per weight than buying cheese in block form, despite being more processed and lower in quality.

Additionally, pre-shredded cheese often contains a significant amount of "cheese powder" – small cheese particles that have been unintentionally separated during the manufacturing, shredding, and packaging process. This cheese "powder" bakes very quickly – much faster than desirable for pizza – making it another reason to avoid pre-shredded cheeses.

"Cheese powder" after a short filtration of pre-shredded cheese

If you still want or need to buy a pre-shredded cheese product, look at the ingredient list for terms like "anti-caking agent," "cellulose," or "starch." If the product contains any of these, it is best to avoid it.

Grande is one of the few cheese manufacturers that do not add anti-caking agents to their pre-shredded cheese products.

If you have already purchased cheese that contains an anti-caking agent and prefer to use it, you can transfer the cheese to a strainer and **rinse it well with water** – the water will help remove the anti-caking agent. After rinsing, dry the cheese thoroughly with a towel or paper towel.

Note that after rinsing, the cheese shreds will tend to stick to each other, so it is best to do this as close to bake time as possible to prevent caking over time.

How to Store Mozzarella and Other Cheeses

Short-Term Storage (1–2 Weeks)

For short-term storage (no more than 1–2 weeks), keep the cheese tightly wrapped in its original packaging in the fridge.

Store it in the coldest part of the fridge, typically the top shelves near the back, where the air vents are located, or in a dedicated cheese or fresh drawer, if your fridge has one.

Long-Term Storage (Over 2 Weeks)

The best way to store cheeses long-term is to **freeze** them.

Most cheeses will maintain their quality well even when frozen **for several months**. However, not all types of cheese retain their quality in the same way, and some are not recommended for freezing (discussed in the next section).

What Happens to Cheese During Freezing

During freezing, the water inside the cheese expands and forms ice crystals, which physically pierce the protein and fat molecules. This **damages the cheese's structure**, potentially altering both its texture and flavor. Once thawed, the cheese can't hold moisture as well, leading to **water loss** and **changes in texture**, often becoming grainy or overly soft.

This change in structure also affects **how the cheese melts and bakes**:

- Lower moisture may cause the cheese to brown more quickly.
- Structural damage may cause more fat to separate during baking, especially in high-fat cheeses.
- The altered structure may reduce meltability and final texture compared to unfrozen cheese.

In general, the **drier** the cheese, the **better** it will tolerate freezing and retain its quality.

For example:

- **Semi-hard cheeses** (like low-moisture mozzarella) and **hard cheeses** (like Parmesan) **freeze very well**, with minimal damage to flavor, texture, or baking behavior.
- **Soft cheeses** (like mascarpone or ricotta) **do not freeze well** – they often become watery or grainy.
- **Fresh mozzarella** *can* be frozen, but it's not ideal; texture damage is likely, resulting in a watery or rubbery consistency after thawing.

How to Freeze Cheese Properly

For optimal quality when freezing cheese, **wrap it tightly in plastic wrap**, place it in a freezer-safe bag, and then freeze. The goal is to minimize air exposure and prevent freezer burn. Alternatively (and preferably), **vacuum seal** the cheese before freezing (which eliminates the need for plastic wrap).

For pizza, it's best to freeze semi-hard cheeses like low-moisture mozzarella in **individual portions sized for one pizza**. Ideally, freeze the cheese **as blocks**, not shredded or sliced. This allows you to thaw only what you need and avoids re-

freezing, which degrades quality.

For example, if you buy a 1–3 kg (2–6 lb) loaf of low-moisture mozzarella, divide it into portions based on how much cheese you typically use per pizza, and freeze them individually. When needed, simply thaw the number of portions required.

To freeze **fresh mozzarella**, remove it from the liquid it's packed in, wrap it tightly in plastic wrap (or vacuum seal it), and freeze. Never freeze it in its original liquid.

How to Use/Thaw Frozen Cheese

To thaw frozen cheese, transfer it from the freezer to the refrigerator **the night before use**.

Slow thawing in the fridge **reduces moisture separation and helps preserve the texture**. While it is possible to thaw cheese at room temperature directly from the freezer, this is not recommended – it increases the risk of texture and flavor degradation.

How to Achieve More Cheese Pull/Stretch

Several key principles and conditions help increase the "cheese pull" effect:

1. Cheese temperature (post-bake)
2. Using the right type of cheese
3. Using quality cheese
4. The way the cheese is cut and applied
5. Amount of cheese used
6. The thickness of the pizza base
7. Temperature, baking time, and avoiding over-baking

1. Cheese Temperature (Post-Bake)

Cheese stretchability and the "cheese pull" effect are **strongly** influenced by the cheese's temperature after baking:

- **Right out of the oven (very hot):** The cheese is fully melted, extensible, and stretches easily – this is when stretchability is at its peak.
- **As it cools:** The melted cheese begins to firm up. Proteins tighten and fat starts to solidify, reducing extensibility and making the cheese less stretchable.

- **When cold:** The cheese becomes very elastic, stiff and brittle, with little to no stretch. Instead of pulling, it tends to snap or break.

In short: Cheese stretches best when hot and freshly melted. As it cools, stretchability steadily declines. While the rate of this change varies by cheese type and brand, the general rule holds for all: **stretch fades as the cheese cools.**

2. Using the Right Type of Cheese

The best cheese for achieving a cheese pull is **whole milk, low-moisture mozzarella (fat content of 24–27%)**. Its low moisture and high fat, along with its protein structure, create an ideal consistency for stretchability after baking.

Avoid pre-shredded cheese, as it contains anti-caking agents that hinder melt and stretch.

Other cheeses with excellent stretchability include: Scamorza, Provolone, Monterey Jack, Brick Cheese, Gruyère.

Cheeses with decent (but not optimal) stretchability: Young/mild Cheddar, Young/mild Gouda, Fontina (best used in combination with the above).

Avoid cheeses with poor stretchability, such as: Fresh mozzarella, Brie, Camembert, aged Cheddar/Gouda, Parmesan, Feta, Mascarpone, Ricotta, long-aged cheeses.

3. Using Quality Cheese

Higher-quality cheese generally melts and stretches better. While specific brands may vary, avoid overly processed or budget cheeses, as they often produce inferior results.

Grande low-moisture mozzarella, for example, is renowned for its excellent melt and stretch.

4. The Way the Cheese Is Cut and Applied

Avoid **large** chunks, strips, or cubes, as they tend to melt into puddles and exhibit reduced stretchability. For maximum cheese pull, it’s best to use **small to medium-sized pieces**, whether shredded, sliced into thin strips, or cubed.

5. Amount of Cheese Used

The more cheese you use, the greater the potential for stretch (provided the cheese is suitable and baking conditions are ideal).

6. The Thickness of the Pizza Base

In general, the **thinner** the pizza base is stretched, the more easily heat transfers from the baking surface through the dough and sauce to the cheese – resulting in **faster cheese baking**. Conversely, a **thicker** base slows down heat transfer, causing the cheese to bake more **slowly**.

So, adjusting the dough thickness directly affects how quickly the cheese bakes.

7. Temperature, Baking Time, and Avoiding Over-Baking

Low baking temperatures (below 270°C / 520°F) and long bake times can cause proteins to seize up and expel fat and water, reducing stretchability.

For best cheese pull results, bake at a high temperature for a short time – this allows the cheese to melt thoroughly without overheating or breaking down its structure, which preserves its stretchability.

What to *Avoid* to Maximize Cheese Pull

- **Fresh mozzarella** (low stretchability)
- **Part-skim, low-moisture mozzarella** (fat < 24%)
- **Over-baking** (damages protein structure and dries the cheese)
- **Aged cheeses** (reduced melt and stretch)
- **Pre-shredded cheese** (contains anti-caking agents that hinder melting/stretch)
- **Letting the cheese sit and cool too much** (stretchability drops quickly as it cools)

Controlling Browning and Oil Separation

It's common for cheese to bake faster or slower than the dough, leading to an unbalanced pizza. The cheese might overbake, brown excessively, or release too much oil before the crust is ready, or, on the flip side, it might still be under-melted when the pizza is done.

There are several ways to adjust the cheese's baking speed (focusing here on the cheese itself, assuming the baking method and other factors including the cheese type remain the same).

How To Slow Down Cheese Baking, Melting and Oil Separation ("Greasing")

Use a **thicker or coarser cut**. The **larger and thicker** the cheese pieces, the **longer** they take to bake. For example:

- For shredded cheese: Use a coarse grater with larger or deeper holes.
- Switching to a different cut/shape: Move from fine shreds to strips or small cubes.
- Use a thicker, chunkier, or coarser cut – thicker pieces take longer to bake, melt, and brown
- Use cold cheese**:** Chill the cheese in the fridge or freezer for about 30 minutes before use. This works better when using thicker cuts. Note that this will only *slightly* delay the cheese's baking.

More methods to slow down the cheese's bake (*not* related to the cheese itself):

- Layering the sauce over the cheese (described in the next section). This acts as a barrier from direct heat and slows down melting and browning.
- Using more cheese: A larger quantity of cheese means more mass and moisture, which takes longer to heat through. This slows melting, delays browning, and can help prevent premature oil separation.

How To Speed Up Cheese Baking

Use a **thinner or finer** cut. **Smaller** pieces bake **faster** and melt more quickly. For example:

- For shredded cheese: Use a finer grater with smaller or shallower holes.
- Thinner strips/cubes: If using strips or cubes, switch to thinner or smaller pieces.

Layering the Sauce Over the Cheese

For detailed information on assembling the sauce over the cheese, see the dedicated article: Tomato Sauce for Pizza: A Complete, Advanced Guide [In-Depth Overview & Practical Tips] (section 'Sauce Layering').

Layering the sauce on top of the cheese serves as an effective way to **slow down the cheese's baking**. Much like oil, the sauce acts as a barrier, shielding the cheese from direct heat and delaying browning.

This technique is especially useful when you want to prevent the cheese from browning excessively.

Mixing Cheese Cuts for Texture and **Appearance**

Using cheese cut into different shapes and thicknesses lets you play with both texture and appearance. For example, combining shredded cheese with chunks, slices, strips, or cubes creates a unique look and mouthfeel on the finished pizza.

Mixing Different Cheeses

Blending multiple types of cheese is an excellent way to create pizzas with unique textures, appearances, and flavors:

- **Texture and Appearance:** As explained earlier, different cheeses behave differently when baked. Combining them can create a pizza with varied melt patterns, color and texture.
- **Flavor:** Since mozzarella has a mild and neutral flavor, pairing it with other cheeses can elevate the overall taste. Whether you mix in soft cheeses (added after baking), semi-soft or semi-hard cheeses (before baking), or hard cheeses (before or after baking), you'll add depth and character to your pizza. This is something I highly recommend experimenting with.

Handling Fresh Mozzarella: Draining Excess Moisture

Fresh mozzarella is high in moisture, which can (and will) **lead to soggy pizza if not handled properly**. It's strongly recommended, if not essential, to **drain the excess moisture before using it**.

To do this:

1. **Cut the cheese**: Cut the mozzarella into your desired shape (cubes, strips, or shreds).
2. **Strain**: Place the pieces in a strainer over a bowl or plate (to catch the draining liquid).
3. **Cover**: To prevent the surface of the cheese from drying out and becoming rubbery, cover the top of the strainer with a towel, plastic wrap, or a plate – anything that shields it from direct air exposure.
4. **Allow to drain**: Let the cheese drain for a few hours before baking.

Semi-moist mozzarella balls like Fior d'Agerola don't need draining, as they naturally contain less moisture. As a general rule, only fresh mozzarella balls stored in liquid require draining.

Using Semi-Hard Cheeses in High-Heat Ovens

While fresh mozzarella is traditionally used for Neapolitan pizzas baked in wood-fired ovens, low-moisture mozzarella and other semi-hard cheeses can also perform well in high-heat settings – **if used correctly**.

Because wood-fired ovens operate at extremely high temperatures, the goal when using drier cheeses is to **delay their baking**, so they have time to melt without overbaking, browning too quickly, or burning.

Here's how to do that:

- **Use thick slices:** Cut the cheese into cubes or thick strips, about 2 cm (1 inch) thick. Avoid using finely shredded cheese, which bakes too quickly. If using shredded cheese is unavoidable, go for coarse/thick shreds.
- **Choose full-fat cheese:** Aim for cheese with a fat content of at least 23% (per 100g). Higher fat helps slow browning and promotes smooth melting.
- **Use high-quality cheese:** Low-grade cheeses tend to brown or burn too fast, especially under high heat.
- **Avoid anti-caking agents:** These will burn quickly and interfere with proper melting.

By following these guidelines, you can get excellent melting and browning results in high-heat applications, even when using drier cheeses like low-moisture mozzarella.

Low-moisture mozzarella strips on a Neapolitan pizza baked in a wood-fired oven

Drizzling Oil Over the Cheese

Drizzling olive oil (or any other oil) over the cheese can help **slow down its baking**. The oil forms a thin barrier between the cheese and the oven's intense heat, effectively shielding it and reducing the risk of early browning.

This technique is especially helpful when using low-moisture mozzarella with a low fat content (< 23%). In such cases, a light drizzle of oil can prevent the cheese from over-browning or browning too quickly. As a bonus, olive oil adds a subtle, rich flavor to the finished pizza.

Keep the drizzle light and even. Using too much oil will make the pizza greasy and unbalanced.

Pizza Cheese: Final Thoughts

Pizza cheese is a vast kingdom, with mozzarella holding the crown, yet many worthy contenders compete, bringing color, depth, and richness to the pizza realm.

As we've seen, a wide variety of cheeses work well on pizza, each offering unique uses and forms. I strongly encourage experimenting with different cheeses, combinations, and application methods based on what's available and your flavor and texture preferences – chances are, this will truly elevate your pizza to the next level.

Key Takeaways

- **Impact on Pizza Quality:**
  The final result is largely dictated by the specific characteristics of the cheese you use, including fat and moisture content, meltability, browning resistance, and oil separation. Experiment with different types and brands to find the cheese that you like best.
- **The Role of Fat Content in Cheese:**
  Fat is the primary driver of both flavor and texture in pizza cheese. It acts as a plasticizer to ensure a smooth, consistent melt. Higher-fat cheeses are typically more flavorful, stay tender longer, and provide a rich "mouthfeel." They also exhibit more oil separation (the classic "orange grease"), which carries and enhances the flavor of the tomato sauce and toppings.
- **Cheese Classifications:**
  - **Fresh/Soft:** high moisture, delicate flavors, spreadable textures – not suitable for baking applications, can be added post-bake (e.g., ricotta).
  - **Semi-Soft:** Creamier than semi-hard cheeses, but firmer than soft cheeses (e.g., blue cheeses).
  - **Semi-Hard:** The "gold standard" for pizza (e.g. mozzarella, cheddar).
  - **Hard/Firm:** Used primarily for flavor and saltiness (e.g., Parmesan).
- **Why Mozzarella Rules:**
  Mozzarella is the "Queen" of pizza cheese due to its unique protein structure, which allows for exceptional melting and stretching. Its neutral flavor profile synergizes with other ingredients rather than overpowering them, making it the ideal base for almost any pizza.
- **Fresh vs. Low-Moisture (Technical):**
  Fresh mozzarella is a high-moisture cheese (60% or more water) characterized by a lower fat and salt content and a very mild, delicate flavor. In contrast, Low-

Moisture mozzarella is denser and more concentrated (45–52% water), resulting in a saltier, higher-fat, and more flavor-intensive profile.

- **Fresh vs. Low-Moisture Mozzarella (Practical):**
  - **Fresh Mozzarella:** Best for a classic Neapolitan appearance where you want distinct white “islands” or “puddles.” It retains its bright white color with minimal oil separation and provides a delicate, milky flavor.
  - **Low-Moisture Mozzarella:** The ideal choice for nearly all other pizza styles. It provides the “classic” pizza experience: it browns well, releases flavorful fats that blend with the sauce, offers high stretchability, and performs excellently when reheated.
- **Quality Between Dairies:**
  Not all mozzarella is equal; different manufacturers use varying fat-to-protein ratios and processing methods. Experimenting with different brands is essential to find the one you like best.
- **Avoid Pre-Shredded Cheese:**
  It is always best to buy cheese in blocks and shred or cut it yourself. Pre-shredded mozzarella contains anti-caking agents (like cellulose) that negatively affect flavor, interfere with melting, and can cause the cheese to brown unevenly or become gritty.
- **Handling Fresh Mozzarella:**
  Because of its high water content, fresh mozzarella should be sliced and drained (or “dried”) before use. This prevents excess moisture from releasing during the bake and making the crust soggy.
- **Experimenting with Application Methods:**
  Don’t be afraid to try different methods. Whether you mix different cheeses together, put the sauce over the cheese instead of under it, or change how you cut it (thick vs. thin or cubes vs. shreds), these changes all affect how the cheese melts, tastes, and blends with the sauce.
- **Storage Tips:**
  Most semi-hard cheeses, including low-moisture mozzarella, can be frozen for long-term storage with minimal loss in quality.